

5.2. Generate the coefficients of an FIR High-Pass filter using Hamming windowing techniques.

```
clc; clear; close all;

%% --- Input Parameters ---

fp = 1000; % High-pass cutoff frequency (Hz)
fs = 8000; % Sampling frequency (Hz)
N = 50; % Filter order
wc = fp/(fs/2); % Normalized cutoff (0–1)

%% --- FIR High Pass using Hamming Window ---
b_hamm = fir1(N, wc, 'high', hamming(N+1));
disp('--- High-Pass Coefficients (Hamming Window) ---');
disp(b_hamm);

%% --- Frequency Response Plots ---

figure;
freqz(b_hamm,1);

title('High-pass FIR using Hamming Window');
title('High-pass FIR using Hanning Window');
```

